

NHANES Summary Report

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Introduction

This report summarizes selected characteristics of the cleaned National Health and Nutrition Examination Survey (NHANES) dataset across the 2015, 2016, and 2017 waves (National Center for Health Statistics, n.d.). The report describes the sample size, gender and ethnicity distributions, and systolic blood pressure measurements for each wave

Sample Size and Gender Distribution

```
sample_gender <- nhanes %>%
  group_by(wave_id) %>%
  summarise(
    Sample_Size = n(),
    Male_N = sum(gender == "Male", na.rm = TRUE),
    Male_Percent = round((Male_N / Sample_Size) * 100, 1)
  )

sample_gender %>%
  kable(
    caption = "Sample Size and Male Distribution by Wave",
    col.names = c("Wave", "Sample Size", "Number of Males", "Male (%)"),
    booktabs = TRUE
  ) %>%
  kable_styling(latex_options = "striped", full_width = FALSE) %>%
  column_spec(1, bold = TRUE)
```

Table 1: Sample Size and Male Distribution by Wave

Wave	Sample Size	Number of Males	Male (%)
2015	10175	5003	49.2
2016	9971	4892	49.1
2017	9254	4557	49.2

The sample size decreased across the three waves, from 10,175 participants in 2015 to 9,254 participants in 2017. The proportion of males remained relatively stable across the study period, ranging from 49.1% to 49.2%.

Ethnicity Distribution

```
ethnicity_summary <- nhanes %>%
  filter(!is.na(ethnicity_2)) %>%
  group_by(wave_id, ethnicity_2) %>%
  summarise(N = n(), .groups = "drop") %>%
  group_by(wave_id) %>%
  mutate(
    Percent = round((N / sum(N)) * 100, 1)
  ) %>%
  ungroup()

ethnicity_summary <- ethnicity_summary %>%
  mutate(
    `N (%)` = paste0(N, " (", Percent, "%)")
  ) %>%
  select(wave_id, ethnicity_2, `N (%)`)

ethnicity_summary %>%
  kable(
    caption = "Ethnicity Distribution by Wave",
    col.names = c("Wave", "Ethnicity", "N (%)"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "repeat_header"),
    full_width = FALSE
  ) %>%
  column_spec(1, bold = TRUE)
```

Table 2: Ethnicity Distribution by Wave

Wave	Ethnicity	N (%)
2015	Asian	1074 (10.6%)
2015	Black	2267 (22.3%)
2015	Mexican	1730 (17%)
2015	Other	470 (4.6%)
2015	Other Hispanic	960 (9.4%)
2015	White	3674 (36.1%)
2016	Asian	1042 (10.5%)
2016	Black	2129 (21.4%)
2016	Mexican	1921 (19.3%)
2016	Other	505 (5.1%)
2016	Other Hispanic	1308 (13.1%)
2016	White	3066 (30.7%)
2017	Asian	1168 (12.6%)
2017	Black	2115 (22.9%)
2017	Mexican	1367 (14.8%)
2017	Other	634 (6.9%)

Table 2: Ethnicity Distribution by Wave (*continued*)

Wave	Ethnicity	N (%)
2017	Other Hispanic	820 (8.9%)
2017	White	3150 (34%)

White participants represented the largest ethnicity category in each wave, accounting for 36.1% of participants in 2015, 30.7% in 2016, and 34.0% in 2017. The distribution of the other ethnicity categories also varied across the three waves

Systolic Blood Pressure

```

nhanes$average_sbp <- rowMeans(
  nhanes[, c("systolic_bp_1",
             "systolic_bp_2",
             "systolic_bp_3",
             "systolic_bp_4")],
  na.rm = TRUE
)

nhanes$average_sbp[is.nan(nhanes$average_sbp)] <- NA

bp_summary <- nhanes %>%
  group_by(wave_id) %>%
  summarise(
    Minimum = min(average_sbp, na.rm = TRUE),
    Maximum = max(average_sbp, na.rm = TRUE),
    Median = median(average_sbp, na.rm = TRUE),
    Mean = mean(average_sbp, na.rm = TRUE),
    SD = sd(average_sbp, na.rm = TRUE)
  )

bp_summary <- bp_summary %>%
  mutate(
    Minimum = round(Minimum, 1),
    Maximum = round(Maximum, 1),
    Median = round(Median, 1),
    Mean = round(Mean, 1),
    SD = round(SD, 1)
  )

bp_summary %>%
  kable(
    caption = "Summary Statistics of Average Systolic Blood Pressure by Wave",
    col.names = c("Wave", "Minimum", "Maximum", "Median", "Mean", "Standard Deviation"),
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = "striped",
    full_width = FALSE
  )

```

```
) %>%  
column_spec(1, bold = TRUE)
```

Table 3: Summary Statistics of Average Systolic Blood Pressure by Wave

Wave	Minimum	Maximum	Median	Mean	Standard Deviation
2015	64.7	228.7	115.3	118.3	18.1
2016	74.0	231.3	117.3	120.4	18.4
2017	72.7	234.0	118.0	121.7	20.3

The mean average systolic blood pressure increased across the three waves, from 118.0 mm Hg in 2015 to 120.0 mm Hg in 2016 and 122.0 mm Hg in 2017. The median showed a similar increase, from 115 mm Hg in 2015 to 118 mm Hg in 2017. The standard deviation ranged from 18.1 to 20.3 mm Hg across the three waves

References

National Center for Health Statistics. n.d. *National Health and Nutrition Examination Survey*. Centers for Disease Control; Prevention. <https://www.cdc.gov/nchs/nhanes/>.